BASE CHANGE COMPATIBILITY OF EXTENSION BY ZERO AND PUSHFORWARD ALONG A FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the identification of the two functors; the closing sentence omits only the verification of functoriality in the sheaf, which is a routine compatibility, and it is preserved verbatim in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Vanishing of the higher direct images of a finite morphism. Same chapter, Proposition proposition-finite-higher-direct-image-zero. Computes the stalks of the pushforward as a finite direct sum of stalks.
- Description of extension by zero and its stalks. Same chapter, Lemma lemma-jshriek-open. Characterizes extension by zero by the vanishing of its stalks outside the open and by its restriction.
- Pushforward commutes with base change for finite morphisms. Same chapter, Lemma lemma-finite-pushforward-commutes-with-base-change. Identifies the restriction of the pushforward with the pushforward of the restriction.

Source reference. Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Navigation pointer invoked at proof line 11634; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F70: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11468–11468. Navigation pointer invoked at proof

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line 11636; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0959: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8292–8292. Navigation pointer invoked at proof line 11642; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} U & \xrightarrow{j'} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ V & \xrightarrow{j} & Y \end{array}$$

where f is finite and j is an open immersion. Then $f_* \circ j'_! = j_! \circ g_*$ as functors $Ab(U_{étale}) \to Ab(Y_{étale})$.

Proof. Let $\mathcal{F}$ be an object of $Ab(U_{étale})$. Let $\overline{y}$ be a geometric point of Y not contained in the open V. Then

$$(f_* j'_! \mathcal{F})_{\overline{y}} = \bigoplus\nolimits_{\overline{x},\ f(\overline{x}) = \overline{y}} (j'_! \mathcal{F})_{\overline{x}} = 0$$

by Proposition 03QP and because the stalk of $j'_!\mathcal{F}$ at $\overline{x} \notin U$ are zero by Lemma 0F70. On the other hand, we have

$$j^{-1} f_* j'_! \mathcal{F} = g_* (j')^{-1} j'_! \mathcal{F} = g_* \mathcal{F}$$

by Lemmas 0959 and 0F70. Hence by the characterization of $j_!$ in Lemma 0F70 we see that $f_* j'_! \mathcal{F} = j_! g_* \mathcal{F}$. We omit the verification that this identification is functorial in $\mathcal{F}$. □